Author: Alper Urk
Subject: Choropleth Mapping Project
Variable: Median Family Income for California Counties (2019–2023)

Motivation

I chose to map median household income across California counties because income distribution reveals a lot about economic opportunity and inequality. California is often seen as a wealthy state, but that wealth isn't evenly distributed. I was curious to see how income varies between urban regions like the Bay Area or Los Angeles and more rural counties.

On a personal level, I've lived in California for the past years and really enjoyed being here. I wanted to better understand where economic opportunities are concentrated. Seeing the income patterns geographically helps me think about which areas are thriving and which may be struggling, and how that might shape both the labor market and quality of life across the state.

Data

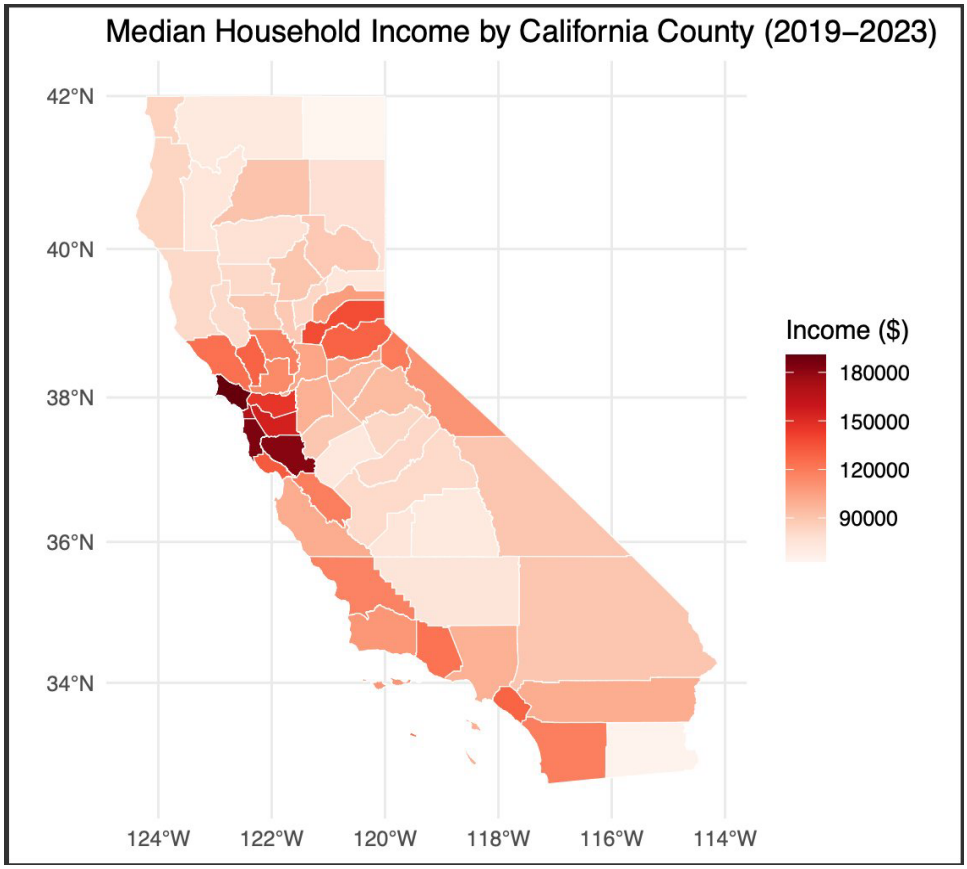
The data comes from the U.S. Department of Health and Human Services' HDPulse dataset, which provides 5-year estimates (2019–2023) of median household income by race, sex, and geography. I focused on the "White (includes Hispanic/Latino), both sexes, all ages" category for consistency across counties. Geographic shapefiles for California counties were obtained from the California State Geoportal. The final dataset included income values for 58 counties in California.

Table 1. Top 10 counties by median household income in California

County	Value (Dollars)
Marin County	190,681
San Mateo County	184,583
Santa Clara County	182,053
San Francisco County	165,603
Alameda County	155,116
Contra Costa County	146,002
Placer County	137,395
Santa Cruz County	131,966
Napa County	129,463
El Dorado County	128,943
Orange County	128,689

Table 2. Bottom 10 counties by median household income in California

County	Value (Dollars)
Modoc County	62,857
Imperial County	64,569
Tulare County	71,096
Siskiyou County	71,583
Merced County	73,010
Sierra County	73,529
Trinity County	73,582
Kings County	74,578
Kern County	75,186
Tehama County	77,022
Lassen County	78,028



Map and Discussion

In the map, counties are shaded based on income; lighter counties have lower median household income, while darker red counties have higher income levels. The wealthiest counties are clustered around the San Francisco Bay Area (Santa Clara, San Mateo, and Marin), where the tech sector has driven up wages significantly. For example, Santa Clara County has a median household income over $159,000. In contrast, inland counties such as Siskiyou, Trinity, and Modoc have income levels under $70,000, showing a clear urban-rural divide.

One interesting and slightly surprising result from the map is that some rural counties near Lake Tahoe, like El Dorado and Nevada counties, show relatively high median household incomes. This is unexpected since rural areas typically have fewer high-paying jobs and lower living costs. However, these counties attract wealthy retirees, remote tech workers, and Bay Area transplants who seek scenic living with access to outdoor recreation. Many own second homes or have moved permanently thanks to remote work flexibility, bringing high incomes into areas with smaller populations.

Additionally, the tourism industry in the Lake Tahoe region plays a key role. High-end lodging, vacation rentals, and services tailored to affluent visitors create jobs and draw in residents with greater spending power. These factors help explain why these counties stand out on the map, rural in geography but increasingly urban in wealth and lifestyle.

Noteworthy Observations

- There is clear income clustering around urban tech hubs in the Bay Area.
- Southern California counties like Orange and San Diego show mid-to-high income, though not as high as Bay Area counties.
- Some counties in Central and Northern California have notably low-income levels despite being in the same state, highlighting internal inequality.